

Professional Summary

Machine Learning Engineer and Statistician with 3+ years of experience in computer vision, time series analysis, and data engineering. Currently working in reinforcement learning and interpretable AI at the Ethiopian Artificial Intelligence Institute, specializing in AI applications for healthcare and finance. Skilled in Python, TensorFlow, and Keras, with experience in building end-to-end data pipelines and developing machine-learning models for medical imaging and financial data. Passionate about leveraging AI to enhance diagnostic accuracy and accessibility in under-resourced settings.

Skills

Technical Skills

- Data Engineering and Analysis: Data Analysis, Data Processing (ETL pipelines, visualization), Time Series Analysis, Sentiment Analysis (VADER), Data Preprocessing
- Model Optimization, Feature Engineering, Python, Streamlit, Ollama Models, API Development, Workflow Automation, Software Development, Data Structures & Algorithms

Machine Learning & AI:

- Computer Vision, Natural Language Processing (NLP), Sentiment Analysis, Model Training, and Fine-Tuning
- Machine Learning, Deep Learning, Computer Vision, Image Classification, Agricultural Image Analysis, AI Recruitment Systems, AI Agents, Large Language Models (LLMs).

Tools

- Python, Streamlit, Ollama, OpenCV, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, FastAPI, Flask, Docker, Git, GitHub, VS Code, Jupyter Notebook, Hugging Face, LangChain, PostgreSQL, SQLite, Google Colab, and MLflow.

Soft Skills

- Problem-Solving, Time Management, Critical Thinking, Communication, and Collaboration.

Work Experience

Machine Learning Engineer,
Ethiopian Artificial Intelligence, Addis Ababa, Ethiopia
Start Date: 2022/23 – Present

Responsibilities & Achievements:

- Developed and deployed AI models for computer vision classification, achieving high accuracy in detecting diseased wheat leaves, and helping farmers take preventive action earlier.
- Built an AI-driven recruitment system, leveraging Streamlit, Ollama models, and AI agents (Extractor, Matcher, Screener, Recommender, Orchestrator), automating candidate screening and reducing manual effort.
- Optimized deep learning pipelines, improving model inference time and scalability, making AI solutions more efficient for real-world deployment.
- Designed and implemented data preprocessing workflows, enhancing data quality and improving training efficiency for machine learning models.
- Collaborated with cross-functional teams to integrate AI solutions into business operations, streamlining workflows and decision-making processes.
- Worked with cutting-edge AI tools such as TensorFlow, PyTorch, OpenCV, FastAPI, and Docker, ensuring robust and scalable AI deployments.

Trainee

10 Academy: Kifiya-AI Mastery Training Program – Ethiopia

- August 2024 – November 2024
- Preprocessed and analyzed diverse datasets, ensuring robustness for model training.
- Fine-tuned deep learning models for tasks like image classification and sentiment analysis.
- Implemented data scraping pipelines for platforms such as Telegram.
- Explored LLMs and explainability frameworks (e.g., SHAP) for advanced analytics.

Projects

- **AI-Powered Wheat Disease Classification**

Description: Developed a computer vision model to classify healthy and diseased wheat leaves using deep learning. Improved model accuracy and inference speed, helping farmers detect crop diseases early and enhance yield protection.

Technologies: TensorFlow, PyTorch, OpenCV, Scikit-learn, Streamlit, FastAPI, Docker.

Link: [\[Github\]](#)

- **AI-Driven Recruitment Workflow**

Description: Built an end-to-end AI-powered recruitment system that automates candidate screening using AI agents (Extractor, Matcher, Screener, Recommender, Orchestrator)—reduced manual effort in hiring and improved matching accuracy.

Technologies: Streamlit, Ollama Models, FastAPI, PostgreSQL, LangChain, Hugging Face, Docker.

Link: [\[Github\]](#)

Education

- **Master of Science in Biostatistics**

College of Natural and Computational Sciences – Hawassa, Ethiopia

October 2021/22 – December 2023

Key Courses: Advanced Computing, Advanced Inference, Bayesian Statistics, and Advanced Machine Learning.

- **Bachelor of Science in Applied Statistics**

College of Natural and Computational Sciences – Dilla, Ethiopia

April 2018 – January 2021

Key Courses: Computing, Programming, Data Science.